



PAPER – 4: COST AND MANAGEMENT ACCOUNTING



QUESTIONS

PART I - Case Scenario based MCQs

Case Scenarios

1. ABC Enterprises is a well-established trading firm specialised in the distribution and sale of a diverse range of products, with a commitment to deliver superior quality to its customers.

Following informations are provided by ABC Enterprises:

- (a) The company is engaged in both cash and credit sales, with credit sales being twice of cash sales. Cash sales are eligible for 3% discount.
- (b) 60% of credit sales are collected within the same month of sale. The remaining 40% is collected in the following month. All receivables are expected to be fully recovered, with no bad debts anticipated.
- (c) The company employs a structured pricing policy, ensuring that the selling price includes a 20% profit margin on the selling price.
- (d) Goods are purchased one month before the month in which these are to be sold.
- (e) The company procures goods through a combination of cash and credit, with cash purchases comprising 40% of the total purchases. Half of the credit purchases are paid within the same month of

purchase, while the remaining 50% is settled in the subsequent month.

- (f) The fourth installment of advance tax, amounting to ₹ 50,000, is due for payment by 15th March, 2025.
- (g) The opening cash in hand as on 1st February, 2025 stands at ₹ 60,000.

The projected sales figures are as follows:

Month	Sales (₹)
January, 2025	9,00,000
February, 2025	7,50,000
March, 2025	8,25,000
April, 2025	6,75,000
May, 2025	6,00,000

On the basis of above Case Scenario, you are required to answer the following MCQs:

- i. What is the net cash sales (after discount) for the month of February 2025?
 - (A) ₹ 2,42,500
 - (B) ₹ 2,50,000
 - (C) ₹ 2,66,750
 - (D) ₹ 2,75,000
- ii. What is the amount of total collection from debtors for the month of March 2025?
 - (A) ₹ 5,13,000
 - (B) ₹ 5,40,000
 - (C) ₹ 5,03,500
 - (D) ₹ 5,30,000

- iii. What is the amount of credit purchases for the month of February 2025?
 - (A) ₹ 3,60,000
 - (B) ₹ 3,96,000
 - (C) ₹ 3,24,000
 - (D) ₹ 4,95,000
 - iv. What is the amount of payment to creditors for the month of March 2025?
 - (A) ₹ 3,42,000
 - (B) ₹ 3,78,000
 - (C) ₹ 3,60,000
 - (D) ₹ 4,50,000
 - v. What is the amount of closing cash in hand as on 31st March, 2025?
 - (A) ₹ 3,61,250
 - (B) ₹ 3,71,250
 - (C) ₹ 3,23,250
 - (D) ₹ 2,00,500
2. White Kollars Pvt. Ltd. is a company engaged solely in the manufacture of shirts, which are bought mainly by office goers. Currently, the company sells its products directly to the retailers. In the recent years, sales has steadily declined due to rising competition. The forecast for next year suggests that the current decline in profits is expected to persist.
- The management believes that a profit of ₹ 60,00,000 is necessary to ensure an adequate return on capital. Thus, it asked the marketing director to review the marketing policies.
- Now, the marketing director submits the following profit and loss account for the last year along with 3 proposals available with him:

(Amount in '000)

Particulars	(₹)	(₹)	(₹)
Sales revenue (50,000 shirts at ₹ 1,500 each)			75,000
Factory cost of goods sold:			
Direct materials	7,500		
Direct labour	26,250		
Variable factory overheads	4,500		
Fixed factory overheads	16,500	54,750	
Fixed Administration overhead		10,500	
Selling and distribution overhead:			
Sales commission (2% of sales)	1,500		
Delivery costs (variable per unit sold)	3,750		
Fixed costs	3,000	8,250	73,500
Profit			1,500

Proposals for upcoming year:

- (i) 10% reduction in selling price which may increase demand by 40%.
- (ii) Selling of 25,000 units to eFlip, an online platform selling products directly to consumers, at a particular price. eFlip would transport the shirts from White Kollars Pvt. Ltd. to its own warehouse. Thus, abolishing the delivery costs and sales commission to be paid for each sale by White Kollars Pvt. Ltd. However, if a suitable price is negotiated, White Kollars Pvt. Ltd. would contribute ₹ 45,00,000 annually towards the online order catalogue cost and provide special packaging at ₹ 75 per shirt.
 Remaining units of 25,000 would continue to be sold at ₹ 1,500 each as per existing business.
- (iii) 10% price reduction together with an advertising campaign costing ₹ 22,50,000 which may boost sales by 60%.

From the information given above, you are required to FIGURE OUT the following for managerial decision:

- i. P/V ratio and Break-even sales (in value) based on the last year accounts would be-
 - (A) 40%, ₹ 7,14,28,571
 - (B) 42%, ₹ 7,14,28,571
 - (C) 40%, ₹ 7,50,00,000
 - (D) 42%, ₹ 7,50,00,000
- ii. Profit/ (Loss) from proposal (i) would be:
 - (A) ₹ 3,38,10,000
 - (B) ₹ 3,00,00,000
 - (C) ₹ 38,10,000
 - (D) ₹ 30,00,000
- iii. With respect to proposal (i), how many number of units White Kollars Pvt. Ltd. would need to sell at ₹ 1,350 each to earn a desired profit of ₹ 60,00,000?
 - (A) 75,000 units
 - (B) 74,534 units
 - (C) 62,112 units
 - (D) 47,619 units
- iv. With respect to proposal (ii), how much selling price per unit be offered to eFlip, that would at least break even on this proposal?
 - (A) ₹ 180
 - (B) ₹ 660
 - (C) ₹ 840
 - (D) ₹ 1,020
- v. Profit/ (Loss) from proposal (iii) would be:
 - (A) ₹ 63,90,000

- (B) ₹ 86,40,000
- (C) ₹ 81,00,000
- (D) (₹ 81,00,000)

3. In two consecutive quarters, a company reported the following performance:

Quarter 1 sales were ₹ 2,10,000 with a profit of ₹ 12,000, while Quarter 2 sales rose to ₹ 2,40,000 with a profit of ₹ 24,000. Assume that the fixed cost remained constant across both quarters and that the company did not introduce any new product lines.

Based on the above information, the P/V ratio must be:

- (A) 20%
- (B) 30%
- (C) 40%
- (D) 50%

4. In a chemical processing unit, materials are added at the beginning of the process. During the month, the department reported 9,200 equivalent units of production (EUP) under the FIFO method.

The process had an opening WIP of 800 units, which were 50% complete, and a closing WIP of 900 units, which were 70% complete with respect to conversion costs.

During the month, the plant also faced a minor machine breakdown, but it did not affect the material input, only the processing speed.

Based on the above information, the number of units completed during the month was:

- (A) 9,100
- (B) 8,970
- (C) 8,450
- (D) 8,650

5. In a manufacturing enterprise, two joint products, X and Y, emerge simultaneously from a common production process. The company incurs joint costs up to the split-off point, after which the products become separately identifiable. For the previous month, the following details relating to production and inventory are available:

Product	Sales (units)	Finished Goods – Opening Stock (units)	Finished Goods – Closing Stock (units)
X	96,000	2,400	6,000
Y	64,000	4,800	2,400

The total joint production cost incurred during the month amounted to ₹ 18,60,000. These joint costs are apportioned between products X and Y in proportion to the number of units produced.

Required:

Determine the amount of joint production cost to be allocated to Product X for the month.

- (A) ₹ 10,23,000
 (B) ₹ 11,49,230
 (C) ₹ 10,71,000
 (D) ₹ 11,16,000

PART - II Descriptive Questions

Material Cost

6. Hari Udyog is a small-scale manufacturing unit producing a single product X using two raw materials PK and BK in the input ratio of 3 : 2. The input–output ratio in production is 5 : 3.

Raw material PK is perishable and becomes obsolete if not used within 5 days of purchase, whereas raw material BK is durable and can be stored for more than one year. Material PK is procured locally with a lead time of 1 to 2 days, while material BK is purchased from a neighbouring state and takes 2 to 4 days to reach the factory.

The estimated sales volume for April 2025 is 30,000 kg, and the same level of sales is expected to continue throughout the year. The company operates for 25 days a month, and production is carried out evenly.

The purchase price per kilogram (excluding taxes) is ₹ 15 for material PK and ₹ 22 for material BK. To place an order, the company incurs an administrative cost of ₹ 120 per order. The annual carrying cost is estimated at 15% of the value for material PK and 5% for material BK.

Currently, material PK is purchased in lots of 8,000 kg to avail a 10% trade discount on the market price. GST at 5% is applicable on material PK, for which input tax credit is available. Material BK is subject to 2% CST, for which no input tax credit is available.

You are required to calculate:

- (i) Economic Order Quantity (EOQ) for each material;
- (ii) Maximum stock level for Material PK;
- (iii) Calculate saving/ loss in Material PK if purchase quantity equals to EOQ.

Employee Cost

7. Rivaan Industries manufactures and sells wooden study tables. The demand for these tables has been steadily increasing due to the company's strong distribution network and reputation for durable products. Despite the rising market potential, the company continues to rely on its long-established wage practices. At present, Rivaan Industries pays wages under a piece-rate system, with a standard piece rate of ₹ 18 per table produced.

The company policy for payment under the piece-rate system is as follows:

- For efficiency less than 100%:
Workers are paid 80% of the standard piece rate
- For efficiency equal to or more than 100%:
Workers are paid 125% of the standard piece rate

The standard output per hour is 3 tables.

Each worker works 9 hours per day from Monday to Friday and 4 hours on Saturday.

A total of 132 workers are currently eligible for this wage plan.

However, the cutting and finishing machines used in production were purchased many years ago and now limit workers' efficiency. As a result, workers have demanded a shift to a time-rate wage system at ₹65 per hour, along with incentives under the Halsey Premium Plan (50% sharing of time saved).

To evaluate the financial impact, the management has prepared the following estimated production levels for January 2026 under three different scenarios:

Production Estimates (January 2026)

Scenario	Worst Case	Optimal Case	Best Case
Production (units)	36,900	73,800	1,10,700

In addition, the company is assessing a potential Technology Upgrade Case involving partial automation, which may influence both output levels and worker productivity.

Required:

- (a) Compute the Total wages, and Average wages per worker per month, for all four scenarios (Worst, Optimal, Best, and Technology Upgrade Case) under each of the following wage systems:
 - (i) Current piece-rate wage system (based on differential piece rates)
 - (ii) Workers' proposed system: Time-rate wages @ ₹ 65 per hour + Halsey Premium (50%)
- (b) Mr. R, one of the workers, produced 900 tables in December 2025. If he produces the same number of tables in January 2026, compute:
 - His earnings under the current piece-rate system
 - His earnings under the proposed time-rate + Halsey incentive system

- The difference in his earnings

Based on your computation, should Mr. R support the workers' demand?

(Assume 4 working weeks in a month.)

Overheads: Absorption Costing method

8. Omega Engineering Pvt. Ltd., a company manufacturing precision components for the automotive sector, operates three major CNC machines in its machining department—Machines X, Y, and Z. Due to a surge in customer demand and rising input costs, the company plans to revise its predetermined machine hour rates for the next financial year. The management has provided the following annual cost data for the three machines:

Preliminary Estimates of Annual Expenses

Particulars	Total (₹)	Machines		
		X (₹)	Y (₹)	Z (₹)
Depreciation	3,00,000	1,20,000	1,00,000	80,000
Spare parts	1,50,000	60,000	50,000	40,000
Power	5,50,000	—	—	—
Consumable stores	1,20,000	50,000	40,000	30,000
Insurance of machinery	1,00,000	—	—	—
Indirect labour	2,40,000	—	—	—
Building maintenance	2,50,000	—	—	—
Annual interest on capital outlay	1,20,000	50,000	40,000	30,000

Additional monthly information:

- Rent & rates: ₹ 30,000 per month
- Salary of Production Supervisor: ₹ 60,000 per month
- Salary of Machine Operator: ₹ 18,000 per month

The supervisor and operator oversee all three machines and spend equal time on each.

The following additional information is also available:

Machine	Estimated Direct Labour Hours	K.W. Rating Ratio	Floor Space (sq. ft.)
X	1,20,000	4	50,000
Y	1,60,000	3	40,000
Z	1,40,000	2	30,000

Working Conditions

- The plant operates 8 hours per day.
- Saturdays are half-days.
- There are 15 holidays besides Sundays, of which 3 fall on Saturdays.
- Machines run at 92% capacity.
- A 3% allowance for breakdown is considered reasonable.

Management expects the following increases for the coming financial year:

1. 10% increase in spare parts prices.
2. 30% increase in spare parts consumption for Machines Y & Z only.

You are required to :

CALCULATE the predetermined machine hour rates for Machines X, Y, and Z after incorporating all the revisions.

Activity Based Costing

9. A XYZ company supplies and maintains a range of printers for business customers. The sales department is responsible for selling the printers and related consumables such as ink and paper. The service department provides after-sales support, including routine preventive maintenance and repair services in the event of equipment breakdown. Customers are charged a fee per print, and this rate varies depending on the type and capacity of the printer.

There is one customer per machine.

The company's existing cost system uses a single overhead rate, based on total sales revenue from print charges, to charge the cost of the service department's support activities to each type of printer.

The service manager has suggested that the print charge should more accurately reflect the costs involved. The company's accountant has decided to implement an activity-based costing (ABC) system and has obtained the following information about the support activities of the service department:

Activity	Cost driver	Overheads per annum ₹ 000
Customer account handling	No. of Customers	126
Planned maintenance Scheduling	No. of planned maintenance visit	480
Unplanned maintenance scheduling	No. of unplanned maintenance visit	147
Spare part procurement	No. of purchase orders	243
Other overheads	No. of machines	600
Total overheads		1,596

The following data have also been collected for each machine size:

	Desktop Inkjet Printers	Office Laser Printers	High-Volume Production colour Printers
Charge per copy	₹ 0.03	₹ 0.04	₹ 0.05
Average number of copies per year per machine	60,000	1,20,000	1,80,000
Number of machines	300	800	500
Planned maintenance visits per machine per year	4	6	12

Unplanned maintenance visits per machine per year	1	1	2
Total number of purchase orders per year	500	1,200	1,000
Cost of parts per maintenance visit	₹ 100	₹ 300	₹ 400
Labour cost per maintenance visit	₹ 60	₹ 80	₹ 100

- Calculate the annual profit per machine for each of the three sizes of machine, using the current basis for charging the costs of support activities to machines.
- Calculate the annual profit per machine for each of the three sizes of machine using activity-based costing (ABC).
- Explain the potential benefits to the company of using an activity-based costing system.

Cost Sheet

- Following information is available from the books of YSPP Ltd. for the current year ending 31st March:

S. No.	Particulars	(₹)	(₹)
(i)	Raw materials purchased		35,00,00,000
(ii)	Freight inwards		39,22,100
(iii)	Wages paid to factory workers		1,02,20,000
(iv)	Contribution made towards employees' PF & ESIS		12,60,000
(v)	Hire charges paid for hiring specific equipment		8,40,000
(vi)	Amount paid for power & fuel		16,17,000
(vii)	Amount paid for purchase of moulds and patterns (life is equivalent to four years production)		31,36,000

(viii)	Job charges paid to job workers		28,42,000
(ix)	Lease rent paid for production assets		3,92,000
(x)	Depreciation on:		
	Factory building	2,94,000	
	Office building	1,96,000	
	Plant & Machinery	4,41,000	
	Delivery vehicles	3,01,000	12,32,000
(xi)	Salary paid to supervisors		4,41,000
(xii)	Repairs & Maintenance paid for:	1,68,000	
	Plant & Machinery		
	Sales office building	63,000	2,31,000
(xiii)	Insurance premium paid for:		
	Plant & Machinery	1,09,200	
	Factory building	63,350	
	Stock of raw materials & WIP	1,26,000	2,98,550
(xiv)	Expenses paid for quality control check activities		68,600
(xv)	Salary paid to quality control staffs		3,36,700
(xvi)	Research & development cost paid for improvement in production process		63,700
(xvii)	Expenses paid for administration of factory work		4,15,100
(xviii)	Salary paid to functional managers:		
	Production control	33,60,000	
	Finance & Accounts	32,13,000	
	Sales & Marketing	35,42,000	1,01,15,000
(xix)	Salary paid to General Manager		43,96,000
(xx)	Packing cost paid for:		
	Primary packing necessary to maintain quality	3,36,000	

	For re-distribution of finished goods	3,92,000	7,28,000
(xxi)	Fee paid to auditors		6,30,000
(xxii)	Fee paid to independent directors		7,70,000
(xxiii)	Value of stock as on 1 st April (beginning):		
	Raw materials	63,00,000	
	Work-in-process	32,20,000	
	Finished goods	38,50,000	1,33,70,000
(xxiv)	Value of stock as on 31 st March (ending):		
	Raw materials	33,60,000	
	Work-in-process	30,45,000	
	Finished goods	63,00,000	1,27,05,000

Due to delay in picking up cargo from the port, YSPP Ltd. had to pay ₹ 15,000 as demurrage in the month of March.

From the above data you are required to PREPARE Statement of cost for YSPP Ltd. for the year ended 31st March, showing (i) Prime cost, (ii) Factory cost, (iii) Cost of Production, (iv) Cost of sales.

Cost Accounting Systems

11. The following figures have been extracted from the cost records of a manufacturing unit:

	(₹)
Stores: Opening balance	32,000
Purchases of material	1,58,000
Transfer from work-in-progress	80,000
Issues to work-in-progress	1,60,000
Issues to repair and maintenance	20,000
Deficiencies found in stock taking	6,000
Work-in-progress: Opening balance	60,000

Direct wages applied	65,000
Overheads applied	2,40,000
Closing balance of W.I.P.	45,000

Finished products: Entire output is sold at a profit of 10% on actual cost from work-in-progress. Wages incurred ₹ 70,000, overhead incurred ₹ 2,50,000.

Items not included in cost records: Income from investment ₹ 10,000, Loss on sale of capital assets ₹ 20,000.

Draw up Store Control account, Work-in-progress Control account, Costing Profit and Loss account, Profit and Loss account and Reconciliation statement.

Job & Batch Costing

12. Analysis of costs by behaviour for decision-making. The Northshire Hospital Trust operates two types of specialist X-ray scanning machine, XR1 and XR50. Details for the next period are estimated as follows:

Machine	XR1	XR50
Running hours	1,100	2,000
	(₹)	(₹)
Variable running costs (excluding plates)	27,500	64,000
Fixed costs	20,000	97,500

A brain scan is normally carried out on machine type XR1: this task uses special X-ray plates costing ₹40 each and takes four hours of machine time. Because of the nature of the process, around 10 per cent of the scans produce blurred and therefore useless results.

Required:

- (a) Calculate the cost of a satisfactory brain scan on machine type XR1.

- (b) Brain scans can also be done on machine type XR50 and would take only 1.8 hours per scan with a reduced reject rate of 6 per cent. However, the cost of the X-ray plates would be ₹ 55 per scan.

Required:

ADVISE which type should be used, assuming sufficient capacity is available on both types of machines.

Process Costing

13. A Company produces a component, which passes through two processes. During the month of April, 2025, materials for 40,000 components were put into Process I of which 30,000 were completed and transferred to Process II. Those not transferred to Process II were 100% complete as to materials cost and 50% complete as to labour and overheads cost. The Process I costs incurred were as follows:

Direct Materials	₹ 15,000
Direct Wages	₹ 18,000
Factory Overheads	₹ 12,000

Of those transferred to Process II, 28,000 units were completed and transferred to finished goods stores. There was a normal loss with no salvage value of 200 units in Process II. There were 1,800 units, remained unfinished in the process with 100% complete as to materials and 25% complete as regard to wages and overheads.

No further process material costs occur after introduction at the first process until the end of the second process, when protective packing is applied to the completed components. The process and packing costs incurred at the end of the Process II were:

Packing Materials	₹ 4,000
Direct Wages	₹ 3,500
Factory Overheads	₹ 4,500

Required:

- (i) PREPARE Statement of Equivalent Production, Cost per unit and Process I A/c.
- (ii) PREPARE statement of Equivalent Production, Cost per unit and Process II A/c.

Joint Product and By-Product

14. ABC Ltd. operates a simple chemical process to convert a single material into three separate items, referred to here as X, Y and Z. All three end products are separated simultaneously at a single split-off point.

Product X and Y are ready for sale immediately upon split off without further processing or any other additional costs. Product Z, however, is processed further before being sold. There is no available market price for Z at the split-off point.

The selling prices quoted here are expected to remain the same in the coming year. During 2024-25, the selling prices of the items and the total amounts sold were:

X – 186 tons sold for ₹ 3,000 per ton

Y – 527 tons sold for ₹ 2,250 per ton

Z – 736 tons sold for ₹ 1,500 per ton

The total joint manufacturing costs for the year were ₹ 12,50,000. An additional ₹ 6,20,000 was spent to finish product Z.

There were no opening inventories of X, Y or Z at the end of the year. The following inventories of complete units were on hand:

X 180 tons

Y 60 Tons

Z 25 tons

There was no opening or closing work-in-progress.

Required:

COMPUTE the cost of inventories of X, Y and Z and cost of goods sold for year ended March 31, 2025, using Net realizable value (NRV) method of joint cost allocation.

Standard Costing

15. BabyMoon Ltd. uses standard costing system in manufacturing one of its product 'Baby Cap'. The details are as follows:

Direct Material 1 Meter @ ₹ 60 per meter	₹ 60
Direct Labour 2 hour @ ₹ 20 per hour	₹ 40
Variable overhead 2 hour @ ₹ 10 per hour	₹ <u>20</u>
Total	₹ <u>120</u>

During the month of August, 10,000 units of 'Baby Cap' were manufactured. Details are as follows:

Direct material consumed	11,400 meters @ ₹ 58 per meter
Direct labour Hours	? @ ? ₹ 4,48,800
Variable overhead incurred	₹ 2,24,400

Variable overhead efficiency variance is ₹ 4,000 A. Variable overheads are based on Direct Labour Hours.

You are required to CALCULATE the following Variances:

- Material Variances- Material Cost Variance, Material Price Variance and Material Usage Variance.
- Variable Overheads variances- Variable overhead Cost Variance, Variable overhead Efficiency Variance and Variable overhead Expenditure Variance.
- Labour variances- Labour Cost Variance, Labour Rate Variance and Labour Efficiency Variance.

Service Costing

16. Fit'n'fine Ltd. runs a fitness club in Delhi. It offers three types of membership plans: Bronze, Silver, and Gold.

The club is using a traditional costing system with the following information:

Cost Information

Description	Amount (in ₹)
Rent	1,00,000 per month
Gym equipments	30,00,000 (service life 5 years)
Licenses & Permits	86,000 (renewed in every six month)
Insurance	12,000 per annum
Salary to the instructors	7,48,000 per annum
Utilities	75,000 per quarter
Cleaning supplies	1,44,000 per annum
AMC of equipments & machinery	1,20,000 per month
Marketing expenses	3,00,000 per annum
Electricity expenses @ ₹ 14 per unit	April to July – 60 units per hour August to March – 25 units per hour

Membership Details

Type of Members	Club Access time	Occupancy % (April to July)	Occupancy % (August to March)
Gold	5.00 AM to 9.00 AM	100%	80%
Silver	6.00 PM to 9.00 PM	70%	50%
Bronze	9.00 AM to 6.00 PM	50%	30%

The maximum capacity of the fitness club is 200 persons in any time slot. The membership fee per month for Gold members is to be fixed at 1.5 times of the Silver membership and that of Silver membership as twice of the Bronze membership fee after providing profit @ 25% on membership fees. The club opens 25 days in a month.

You are required to calculate the membership fee per month to be charged by the club for each member type.

Miscellaneous

17. (a) A manufacturing firm is planning to introduce a new product while simultaneously considering changes in its production methods, advertising budget, and investment plans. As the management accountant, you are required to advise the management on how different types of costs should be treated for effective decision-making. Explain how pre-determined cost, Differential Cost, product cost, out-of-pocket cost, shutdown cost, sunk cost, and implicit cost would influence managerial decisions in this scenario.
- (b) A furniture manufacturing company produces wooden tables. During production in June, the supervisor reports two issues:
- Some wooden pieces get completely damaged during cutting and can only be sold as low-value waste.
 - A few finished tables have minor defects like rough edges or uneven legs, which can be corrected with extra labour and polishing.

As the cost accountant, explain:

1. How scrap differs from defective products in this scenario.
2. accounting treatment of scrap and defectives in the cost accounts.